

DESIGN PROJECT REPORT

BenMedHallu: A Fact-Controlled Benchmark for Hallucination Detection in Bengali Medical Text

CSE 4610 : Design Project

Nashat Islam

220041211

Subha Tamanna Mrittika

220041227

Atia Zaman

220041241

Supervised by:

Ajwad Abrar Mostofa, Junior Lecturer, Department of Computer Science and
Engineering

Tareque Mohmud Chowdhury, Assistant Professor, Department of Computer
Science and Engineering

Department of Computer Science and Engineering

Islamic University of Technology

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Declaration of Candidate

This is to certify that the work presented in this thesis is the outcome of the analysis and experiments carried out by **Nashat Islam**, **Subha Tamanna Mrittika**, and **Atia Zaman** under the supervision of **Ajwad Abrar Mostofa**, Junior Lecturer, Department of Computer Science and Engineering and co-supervision of **Tareque Mohmud Chowdhury**, Assistant Professor, Department of Computer Science and Engineering, Islamic University of Technology, Dhaka, Bangladesh. It is also declared that neither this thesis nor any part of it has been submitted anywhere else for any degree or diploma. Information derived from the published and unpublished work of others have been acknowledged in the text and a list of references is given.

Ajwad Abrar Mostofa

Junior Lecturer

Department of Computer Science and Engineering

Islamic University of Technology (IUT)

Date: September 14, 2026

Nashat Islam

Student ID: 220041211

Date: September 14, 2026

Subha Tamanna Mrittika

Student ID: 220041227

Date: September 14, 2026

Tareque Mohmud Chowdhury

Assistant Professor

Department of Computer Science and Engineering

Islamic University of Technology (IUT)

Date: September 14, 2026

Atia Zaman

Student ID: 220041241

Date: September 14, 2026

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List of Abbreviations

API	Application Programming Interface
BIO	Beginning, Inside, Outside (sequence tagging scheme)
CHQ	Consumer Health Question
LLM	Large Language Model
MCQ	Multiple Choice Question
NER	Named Entity Recognition
NLP	Natural Language Processing
QA	Question Answering
Q-ALT	Question-Alteration track of this benchmark
RNG	Random Number Generator

Table 1: Marks Distribution Summary

CO No.	Course Outcome	Contributig Chapters	Marks
CO1	Identify research problems and implement innovative solutions by exploring, analyzing, and applying state-of-the-art technologies, frameworks, tools, and algorithms to address complex real-world engineering challenges.	Chapter 1, 2 and 3	10
CO2	Demonstrate effective teamwork, professional ethics, and responsibility by applying project management principles to plan, execute, and deliver a complex software system within defined constraints.	Appendix A and B	5
CO3	Evaluate and design computing solutions considering ethical, societal, and environmental implications, ensuring responsible and sustainable engineering practices.	Chapter 4, 5 and 7	10
CO5	Build a strong foundation for future research in the same or related domains.	Chapter 8 and 9	5

Abstract

Large language models increasingly answer health questions in Bengali, a language spoken by more than 230 million people but poorly served by clinical language resources. When such a model substitutes one medicine for another or softens a reported test value, the output stays fluent and confident, and nothing on the surface marks it as wrong. No benchmark measured whether a model can separate faithful Bengali medical text from corrupted text. This project builds one. Rather than annotating hallucinations found in model output, we manufacture them: authentic Bengali clinical text is edited under programmatic control, so the gold label is an output of the construction pipeline rather than a human judgement about it. The benchmark spans four task formats — entity swapping, answer pairing, question-premise corruption, and source corruption for summarization — yielding 19,396 labelled items and 104,502 detector judgements across six large language models. The results show a sharp asymmetry: detectors reach 94–100% accuracy on faithful text but between 11% and 95% recall on corrupted text. Failure is structured, tracking both the clinical entity type swapped and the corruption operation applied, which makes the benchmark diagnostic rather than merely comparative.

Chapter 1

Introduction

1.1 Background and Motivation

A *hallucination* in natural language generation is output that is fluent and plausible but unsupported by the source or by fact [7]. It covers two failures that behave differently under measurement: an *extrinsic* hallucination asserts something the source never said, an *intrinsic* one something the source contradicts [14].

Large language models already answer health questions for Bengali speakers, a population of more than 230 million. Here a hallucination is not a stylistic defect. A model that swaps a drug name, changes a dose, or drops a laboratory value produces text that remains grammatical, confident and recognisably clinical. The error is invisible at the surface, and the reader most exposed to it is least equipped to catch it. Three edits drawn from the benchmark built in this project, shown in Table 1.1, illustrate the point:

Table 1.1: Corruptions that leave the text fluent. Each is an item from this benchmark.

Original	Corrupted	Nature of the edit
হিস্টাসিন	সিটিরিজিন	A different drug of the same class
৫.৯ ইঞ্চি	৫.৯ ফুট	The unit is rescaled
রেজাল্ট পজিটিভ আসছে	রেজাল্ট ভালো আসছে	A specific result is generalised away

Bengali compounds the problem. It is low-resource for clinical NLP, with no corpus of annotated medical faithfulness judgements, and the text itself is difficult: patient messages are informal, code-mixed with English medical terms, inconsis-

tently spelled, and often repeat one clinical detail in several paraphrases. Methods validated on English clinical notes have little purchase here. Hallucination benchmarking is nonetheless mature — in English. TruthfulQA [12], HaluEval [11] and FactCC [10] are English resources; Med-HALT [16] and MedHallu [17] add the medical domain but stay English-centric. For Bengali medical text the question had never been asked in a measurable way.

1.2 Problem Statement and Objectives

Problem statement. No resource measures whether large language models can distinguish faithful Bengali medical text from deliberately corrupted text, and no methodology exists for building one in which the gold label, the error type and the item difficulty are all under the builder’s control.

The project pursues four objectives:

1. Design a construction methodology in which hallucinated items are manufactured by applying a single named, recorded edit to authentic Bengali medical text.
2. Instantiate that methodology across four task formats — entity, answer, question and document level.
3. Evaluate six large language models as detectors under one protocol, reporting accuracy by entity type, corruption operation and cost.
4. Keep the models that generate the data and the models evaluated on it in disjoint families.

1.3 Research Challenges

Four challenges are specific to this problem. *Distribution leakage*: if hallucinated items are machine-written and faithful ones human-written, a detector can win by recognising generated prose rather than by checking facts. *Label certainty*: an edit meant to remove support may fail silently, leaving an item mislabelled and penalising every detector that answers correctly. *Register preservation*: patients repeat a detail in several forms, so a partial edit leaves the document contradicting itself — a different phenomenon from the one being built. *Attributable difficulty*: for a score to be diagnostic rather than merely low, exactly one experimental axis may move at a time.

1.4 Contributions

This project delivers a hallucination-detection benchmark for Bengali medical text covering four task formats and 19,396 labelled items; a construction method that derives gold labels from the pipeline and records the changed span with every item; a novel corruption operation, *Coarsen*, which isolates partial support and proves the hardest condition in the benchmark; an evaluation of six detectors disaggregated by entity type, operation and cost; and a generator/detector separation that keeps the benchmark outside the self-evaluation loop prior Bengali work identifies as its own limitation.

1.5 Justification as a Complex Engineering Problem

The problem has no codified solution: constructing labelled hallucinations in a low-resource language required designing, piloting and empirically pruning six candidate corruption operations. It involves conflicting requirements, since edits must be semantically decisive yet linguistically invisible, and demands domain knowledge spanning clinical semantics, Bengali sociolinguistics and distributed system reliability. Its outcome affects public health information access for over 230 million speakers.

1.6 Organisation of the Report

chapter 2 surveys related work and identifies three gaps. chapter 3 states the research questions and the scope. chapter 4 presents the construction methodology and compares candidate detection approaches. chapter 5 describes the system as built. chapter 6 reports results for all four tracks. chapter 7 addresses ethical, social and environmental considerations, chapter 8 the challenges and limitations, and chapter 9 concludes.

Chapter 2

Related Works

2.1 Two Ways to Build a Benchmark

One family of benchmarks fixes a task, collects model output, and labels the errors that appear. TruthfulQA [12] curates adversarial questions on which models reproduce human misconceptions; HaluEval [11] scales this by sampling responses and filtering them through human review. Both share a structural property: the error types present are whatever the generating model happened to produce, and item difficulty is unknown to the builder. Neither can answer whether detection degrades for one corruption type rather than another, because type was never a controlled variable.

The second family manufactures errors with known properties. FactCC [10] applies rule-based transformations to reference summaries, gaining control at the cost of transformations superficial enough for a classifier to learn the artefact rather than the semantics. BN [4] go further for clinical summarization with a Leave- N -Out procedure on English clinical dialogue from ACI-Bench [19]: atomic facts are extracted from a reference, N are removed from the input, and N becomes an explicit difficulty dial rather than a post-hoc bin. This is the closest ancestor of the present work, though ACI-Bench holds structured English encounters whereas our corpus holds unstructured Bengali patient messages. FActScore [15] decomposes text into atomic facts; we reuse the idea but apply it to the *reference*, in order to choose what to corrupt.

2.2 Medical Hallucination Beyond English

Med-HALT [16] brought hallucination testing into medicine through examination items drawn from several countries, but frames its tasks in English and treats hallucination as incorrect recall rather than unfaithfulness to a source. MedHallu [17] is closer in method to this work: it builds 10,000 medical question–answer pairs on PubMedQA, of which 1,000 are human-annotated and 9,000 deliberately generated, and stratifies them into difficulty tiers. Its finding that domain specificity amplifies the impact of hallucination motivates a medical benchmark rather than a general one, but its text is English and journal-derived, far from the informal patient register studied here.

CMHE [5] is the clearest precedent for a non-English medical benchmark: 42,000 items built from Chinese medical dialogue, covering detection, diagnosis and explanation, with a portion re-verified by medical professionals. It establishes that an expert-validated medical hallucination benchmark is feasible outside English — and, by doing so for Chinese, sharpens the absence of any equivalent for Bengali.

2.3 Detection Methods and Bengali Resources

Detection approaches divide into three. Trained entailment classifiers [10] are cheap at inference but need labelled training data in the target language. Sampling-consistency methods such as SelfCheckGPT [13] need neither reference nor training data, detecting hallucination through disagreement across stochastic samples, at the cost of several generations per item. Zero-shot prompting of a strong model — the LLM-as-a-judge paradigm analysed by Zheng et al. [20] — needs neither, but inherits the judge’s biases. section 4.5 returns to this choice with explicit criteria.

For Bengali the available resources are largely general-domain: BanglaBERT [3] supplies pretrained representations and XL-Sum [6] news summarization data. Two recent works are closer. BenHalluEval [1] constructs the first dedicated hallucination benchmark for Bengali, spanning question answering, code-mixed question answering, summarization and mathematical reasoning, with 12,000 hallucinated candidates evaluated under a dual-track protocol that scores ground-truth and hallucinated items separately. We adopt that dual-track protocol directly. Where BenHalluEval corrupts the *summary*, the present work corrupts the *source* and never edits the human summary, and it narrows the domain from general Bengali to Bengali clinical text. BanglaSummEval [2] applies LLM judges to Bengali summarization output and explicitly names the self-reinforcing loop that arises when judge

and evaluated models share a family as a limitation of its own design.

2.4 Identified Gaps

Table 2.1 positions this work. The *linguistic* gap: medical hallucination benchmarks exist for English and for Chinese, while the Bengali entries are general-domain. The *label provenance* gap: annotation-based benchmarks label errors they encountered rather than errors they designed, leaving type, difficulty and balance uncontrolled; only the construction-based line escapes this, and it had not been applied to a low-resource language. The *circularity* gap: where generator, detector and judge share a family, agreement within that family is indistinguishable from faithfulness.

Table 2.1: Positioning of related work along task, language and label provenance

Work	Task	Language	Label provenance
FactCC [10]	Summarization consistency	English	Rule-based synthesis
TruthfulQA [12]	QA truthfulness	English	Human curation
HaluEval [11]	QA, dialogue, summ.	English	Sampling + human filter
FActScore [15]	Atomic factual precision	English	Retrieval + LM validation
SelfCheckGPT [13]	Detection method	English	Sampling consistency
Med-HALT [16]	Medical reasoning	English-centric	Exam-derived
MedHallu [17]	Medical QA	English	1k human + 9k generated
CMHE [5]	Medical detection	Chinese	Generated + expert re-check
Leave- <i>N</i> -Out [4]	Clinical summarization	English	By construction
BenHalluEval [1]	Four tasks, general	Bengali	Generated + 3 annotators
BanglaSummEval [2]	Summarization eval.	Bengali	LLM judges
BenMedHallu (this work)	Four formats, medical	Bengali	By construction

Chapter 3

Problem Definition and Scope

3.1 Problem and Research Questions

Given a piece of Bengali medical text and its reference — a tagged clinical entity and its sentence, a question and a proposed answer, or a source document and its human-written summary — can a contemporary large language model determine whether the text is faithful to that reference, and on what does its ability depend? The problem is unsolved on three counts: no labelled Bengali medical faithfulness data exists, so the question cannot be asked; no construction methodology exists for such data in a low-resource setting, so the data cannot easily be made; and no protocol exists that would make a resulting score diagnostic rather than merely comparative.

RQ1 Can contemporary LLMs detect entity-level hallucination in Bengali clinical text at all?

RQ2 Does detection depend on *what* was corrupted — the clinical entity type?

RQ3 Does it depend on *how* — deletion, substitution, or coarsening?

RQ4 Do model scale and price predict detection ability?

3.2 Scope and Delimitations

Within one semester the project builds four task formats, evaluates six detectors under one protocol, and disaggregates results by entity type, operation and cost. Explicitly out of scope: no model is fine-tuned; no hallucination mitigation or repair is attempted; no clinical validation by practising physicians is performed; only binary

detection is scored, with edit spans recorded but not scored for localisation; and the summarization track is scored on its test and validation splits only.

Chapter 4

Proposed Methodology

4.1 Overview

The methodology rests on one inversion. Instead of generating text and labelling the hallucinations in it, we take authentic Bengali medical text known to be faithful and apply exactly one controlled, recorded edit that destroys that faithfulness. The label is an output of the pipeline rather than a judgement about it, and the record of what changed comes free. Figure 4.1 shows the complete system.

Three properties make later results interpretable. *One text distribution*: because the edit is applied to authentic text and the reference is never rewritten, faithful and hallucinated items are drawn from the same distribution, so a detector cannot win by recognising machine-written prose. *Selection in code*: the target of each edit is chosen by a seeded random number generator inside the pipeline, not by the generating model. Letting the model choose would cost the gold record of what changed, the balance across categories, and the difficulty dial, and would make two runs of the same pipeline produce different data. *One axis at a time*: the deletion pipeline holds the operation fixed and varies the number of facts removed, while the alteration pipeline holds that number at one and varies the method, so a drop in accuracy is always attributable.

4.2 The Four Tracks

Track 1 (NER) takes a Bengali clinical sentence with tagged entities, converted from BIO tagging into one column per entity type, and replaces exactly one tagged entity with a different entity of the same class. The prompt requires the replacement

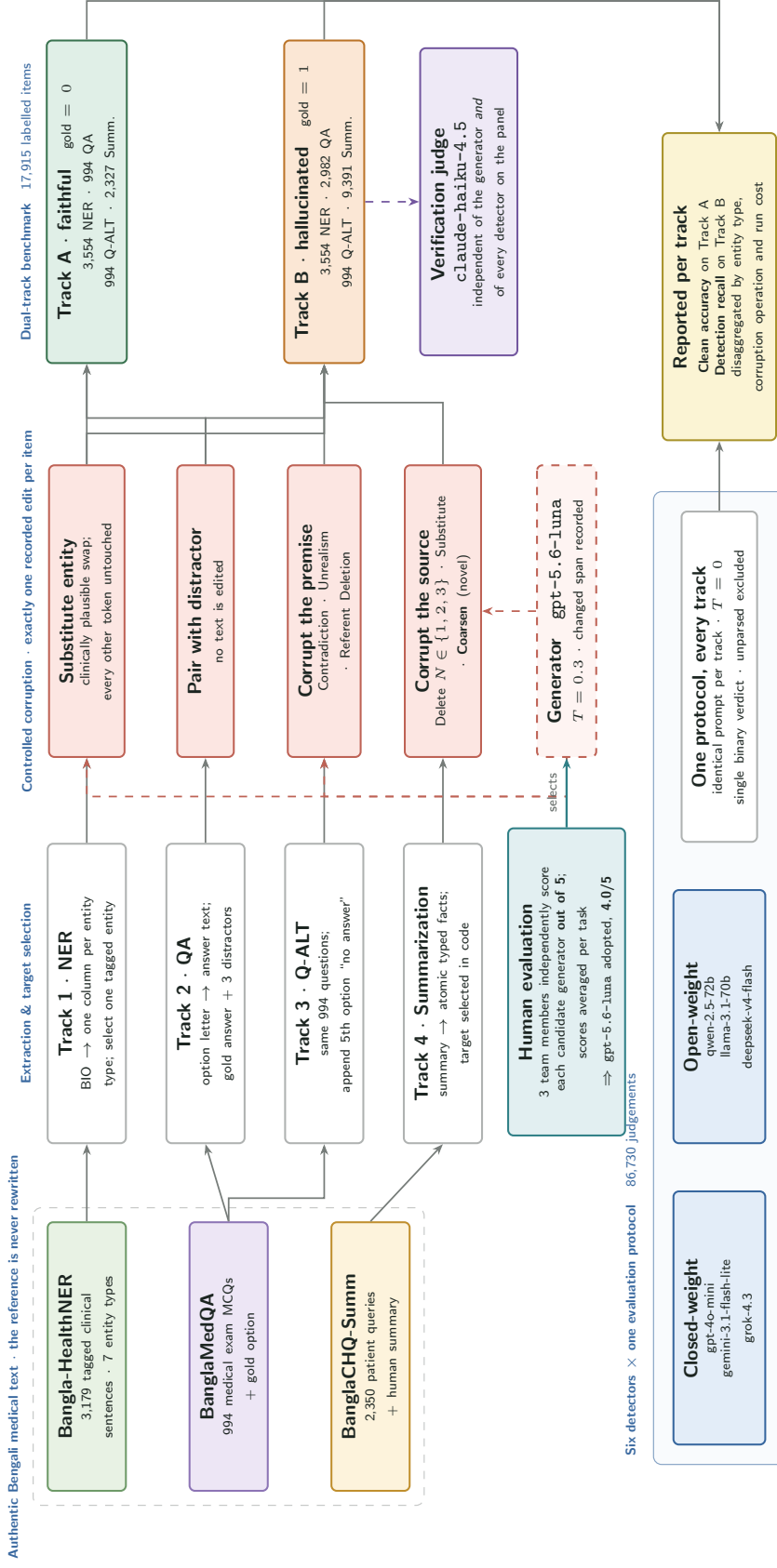


Figure 4.1: End-to-end construction and evaluation pipeline. Three authentic corpora feed four tracks; each track applies exactly one recorded corruption; every item lands in Track A (faithful) or Track B (hallucinated); six detectors score every item under one protocol.

to be clinically plausible, different from the original, absent elsewhere in the sentence, and to leave every other token untouched. The detector sees the sentence and the target entity and returns one binary digit. Each row is judged twice, clean and corrupted.

Track 2 (QA) takes a Bengali medical examination question with four options. No text is corrupted at all: the option letter is resolved to its full answer text, and the question is paired once with the correct answer and once with each of the three distractors written by the examiners. Because nothing is generated, this track acts as a control on the rest of the benchmark, and the agreement between the three distractor variants indicates whether the measurement is stable.

Track 3 (Q-ALT) uses the same 994 questions but corrupts the question rather than the answer. Each is rewritten so that none of options A–D can be correct, and a fifth option, “উত্তর নেই” (“no answer”), is added. Three techniques are used. *Contradiction* embeds two individually plausible but jointly impossible premises. *Unrealism* states a clinically styled but factually wrong value or property. *Referent Deletion* removes the essential detail behind a confident anaphoric reference whose antecedent was never supplied. Only selecting the fifth option counts as detection.

Track 4 (Summarization) carries the central inversion. Conventionally the source is held fixed and the summary rewritten, which makes every hallucinated summary machine-written and every faithful one human-written. Here the human summary is never edited; the *source* is rewritten until it no longer supports that summary. Four stages follow: the summary is decomposed into atomic, typed facts across ten clinical categories; the target is selected in code from a seeded RNG; the generator rewrites the source so the target loses support while retained facts survive; and an independent judge verifies each fact before acceptance.

Category balance is enforced during selection. Symptom accounts for 47% of all extracted facts, which would dominate the benchmark if sampled proportionally, so it is capped at one target per document; after the cap it accounts for 26.7% of targets. Two categories are excluded as alteration targets on pilot evidence: Specialist, which appeared once across fifty pilot documents, and Medical Procedure, roughly half of whose instances were mislabelled in the source corpus. The Other category has identifiable spans but no swappable value slot, so it is targeted by deletion only.

4.3 Selecting the Corruption Operations

Six candidate operations were piloted on fifty documents; three shipped and three were cut, on measurement rather than judgement. The metric was *claim survival*: the proportion of what the source originally said that is still present verbatim after the edit. An operation that removes nothing leaves the claim recoverable, which makes the gold label arguable. Deletion removed the most support at 21% survival, followed by Substitute at 60% and Coarsen at 65%; the three cut operations left far more intact — Reattribute 96%, Weaken 97% and Conflict 100%. All three read convincingly by hand, and only the measurement exposed them.

The survivors produce three distinct support relations. **Delete** N , for $N \in \{1, 2, 3\}$, removes N atomic facts, leaving claims *unsupported* — the extrinsic case. **Substitute** swaps a value for a plausible clinical neighbour, leaving the claim *contradicted* — the intrinsic case, and the arm comparable to FactCC. **Coarsen** replaces a specific value with a vaguer one that remains true of the original: where a source reported that a test kept coming back **নেগেটিভ** (“negative”), the coarsened source says it kept **আশানুরূপ হচ্ছে না** (“not turning out as hoped”). The summary’s specific value is then unsupported while its direction of travel still is. This *partial* support is isolated by no existing benchmark we are aware of, and proves the hardest case here.

The two pipelines stay separate because they compound differently. Removing three facts is a harder version of removing one, so N works as a difficulty dial; altering three values at once produces a document rewritten in three unrelated ways and a failure attributable to none of them.

4.4 Verification and Anti-Circularity

Verification checks the *data*; the detector evaluation measures the *models*. The order matters: if a deletion silently failed and the summary is still supported, the item is labelled hallucinated, every detector correctly reports faithful, and all six are penalised for being right.

The design uses one batched call per item. The judge is shown the corrupted source together with every summary fact — targeted and retained, shuffled so that position leaks nothing — and returns a verdict per line. For deletion, every targeted fact must be NOT_MENTIONED; for alteration, CONTRADICTED; for both, every retained fact must remain SUPPORTED. The judge must be independent: not the generator, which would rubber-stamp its own edits, and not a member of the detector panel,

which would make the benchmark circular.

This stage was implemented and piloted on 400 items but was not run over the full assembled dataset within the project timeline; chapter 8 treats the consequence as the most significant limitation of the work. Two further filters need no model calls and were applied to the whole dataset: any item whose corrupted source came back byte-identical to the original is dropped, as is any deletion that did not shrink the document.

4.5 Comparison of State-of-the-Art Approaches

Three established approaches could serve as the detection component; Table 4.1 compares them against the criteria that matter here.

Table 4.1: Candidate detection approaches against project selection criteria

Criterion	Trained classifier	Sampling consistency	Zero-shot judge
Bengali training data	Required	Not required	Not required
Uses a reference	Yes	No	Yes
Calls per item	1 (local)	5–20	1
Relative cost	Lowest	Highest	Low
Cross-model comparison	Poor	Moderate	Direct
Reproducibility	High	Low (stochastic)	High at $T = 0$
Main weakness	No Bengali data	Proxies self-consistency	Inherits judge bias

The **trained classifier** is cheapest at inference and most reproducible, but is ruled out by the very gap this project exists to close: there is no labelled Bengali medical faithfulness corpus to train on, so building the benchmark is a precondition for this approach rather than an alternative to it. **Sampling consistency** needs no training data, but measures whether a model agrees with itself — a proxy for hallucination rather than a measurement of it — is inapplicable where a reference exists and ought to be used, and costs five to twenty generations per item.

The **zero-shot LLM judge** was adopted. The decisive criteria were that no Bengali training data is needed; that the reference is available and used; that one call per item keeps the benchmark affordable; and, most importantly for a benchmark built for comparison, that an identical prompt can be issued to every candidate model, so score differences reflect models rather than detection pipelines. Its known weakness is mitigated structurally by keeping the generating model outside the detector panel.

Chapter 5

Implementation

5.1 Architecture and Configuration

The repository mirrors a four-stage flow: `preprocessing/` holds the generation pipelines, `dataset/` the assembled benchmark, the `evaluation_*` trees the raw detector runs, and aggregation scripts convert verdicts into the reported tables. Prompts and selection logic live in importable modules rather than notebooks, so that each prompt exists in exactly one place: `common.py`, `b1_deletion.py`, `b2_alteration.py` and `verification.py`.

Every track runs the same six stages, and only two of them differ between tracks. Table 5.1 sets them side by side.

Table 5.1: One pipeline, four tracks. Only extraction and corruption differ; selection, filtering, evaluation and aggregation are the same code path throughout.

Stage	T1 NER	T2 QA	T3 Q-ALT	T4 Summ.
1. Extract	BIOES to entity columns	option letter to answer text	append a fifth option	summary to atomic typed facts
2. Select	a seeded RNG chooses the target in code, not the model			
3. Generate	swap the entity	<i>none</i>	rewrite the premise	rewrite the source
4. Filter	drop unchanged items; drop deletions that did not shrink the text			
5. Evaluate	six detectors, identical prompt per track, $T = 0$, one binary verdict			
6. Aggregate	verdicts joined by item id into per-category and per-operation tables			

All models are reached through OpenRouter, an OpenAI-compatible gateway, giving one client, one retry policy and one cost ledger across six vendors. The generator is `gpt-5.6-luna`; the detectors are `gpt-4o-mini`, `gemini-3.1-flash-lite` and `grok-4.3` (closed-weight) and `qwen-2.5-72b-instruct`, `llama-3.1-70b-instruct` and `deepseek-v4-flash` (open-weight). Generation runs at temperature 0.3 and every evaluation call at 0.0. Completion-token caps are set per prompt — three tokens on the binary NER prompt, but well above the reasoning budget on the summarization prompt, since a reasoning-capable model whose budget exceeds the cap returns an empty completion rather than an error.

5.2 Source Corpora

Three public corpora supply the authentic text, summarised in Table 5.2. All three are Bengali and medical, and none is rewritten at any stage.

Table 5.2: Source corpora. The reference in each is treated as ground truth and never edited.

Corpus	Content	Used for
Bangla-HealthNER [9]	tagged clinical sentences	Track 1
BanglaMedQA [18]	medical examination MCQs	Tracks 2 and 3
BanglaCHQ-Summ [8]	patient queries with human summaries	Track 4

5.3 Generator Selection

The generator choice was measured rather than assumed. Four candidate models produced corruptions for a pilot, and the three project members independently rated each candidate’s output out of five for every task, with the scores averaged. Table 5.3 gives the result. `gpt-5.6-luna` led or tied on every task and was adopted as the sole generator; the other three were not used to produce any released item.

Table 5.3: Generator selection by human evaluation. Mean rating out of 5 over three independent reviewers.

Candidate generator	Medicine	Symptom	Health Cond.	Summarization	Mean
<code>gpt-5.6-luna</code>	4.0	4.0	4.0	4.0	4.00
<code>deepseek-v4-pro</code>	4.0	4.0	3.5	3.0	3.63
<code>claude-opus-4.6</code>	3.0	4.0	3.5	3.5	3.50
<code>qwen-3.8-max</code>	3.5	3.0	3.0	3.5	3.25

5.4 Determinism and Reliability

Selection is seeded per document, so reruns are reproducible and targets at $N=2$ are a strict superset of the $N=1$ target for the same document, making a change in accuracy attributable to the facts added rather than to a different draw.

Runs against six commercial APIs at tens of thousands of calls fail constantly, and four mechanisms were required to complete them. Results are checkpointed every fifty rows and every run resumes from its own output file, so an interruption never recomputes completed work. Multiple API keys rotate on HTTP 402 and 429. A repair pass re-runs only those rows whose stored response matches a connection-error pattern or whose verdict is null. A pre-flight probe issues one call per detector before committing a full run, a safeguard added after a reasoning model silently returned empty completions for a batch.

5.5 The Resulting Benchmark

Table 5.4 gives the assembled benchmark. Judgement counts are actual rather than nominal: every item is judged by all six detectors, and NER items are judged twice, once clean and once corrupted.

Table 5.4: Benchmark inventory. NER pairs are judged twice, clean and corrupted.

Track	Source material	Labelled items	Judgements
NER	4,660 tagged clinical sentences	5,035 pairs	60,420
QA	994 medical exam questions	3,976 pairs	23,856
Q-ALT	The same 994 questions, rewritten	994 items	5,964
Summarization	2,327 messages with human summaries	9,391 items	14,262
Total		19,396	104,502

The summarization corpus splits 1,858 / 234 / 235 across train, validation and test, and corruption yields 9,391 accepted items: Substitute 2,305, Delete $N=1$ 2,240, Delete $N=2$ 1,942, Coarsen 1,465 and Delete $N=3$ 1,439. The NER test set covers seven entity types, of which six were evaluated: Medicine (1,058 sentences), Specialist (650), Health Condition (577), Age (532), Dosage (495) and Medical Procedure (242). Symptom, at 1,481 sentences the largest, reached generation only.

Chapter 6

Results and Discussion

All results use temperature 0.0 and one prompt per track, issued identically to each detector. NER items are judged in pairs, so a detector’s clean and corrupted figures are computed over the same underlying text. The summarization track covers the test and validation splits, giving 2,377 items per detector. Unparseable verdicts are excluded from the denominator rather than defaulted to a value. Throughout, *clean accuracy* means accuracy on the faithful member of a pair and *recall* the proportion of corrupted items correctly flagged; the gap between them is the principal finding of this project.

6.1 Overview

Figure 6.1 places every detector on every track.

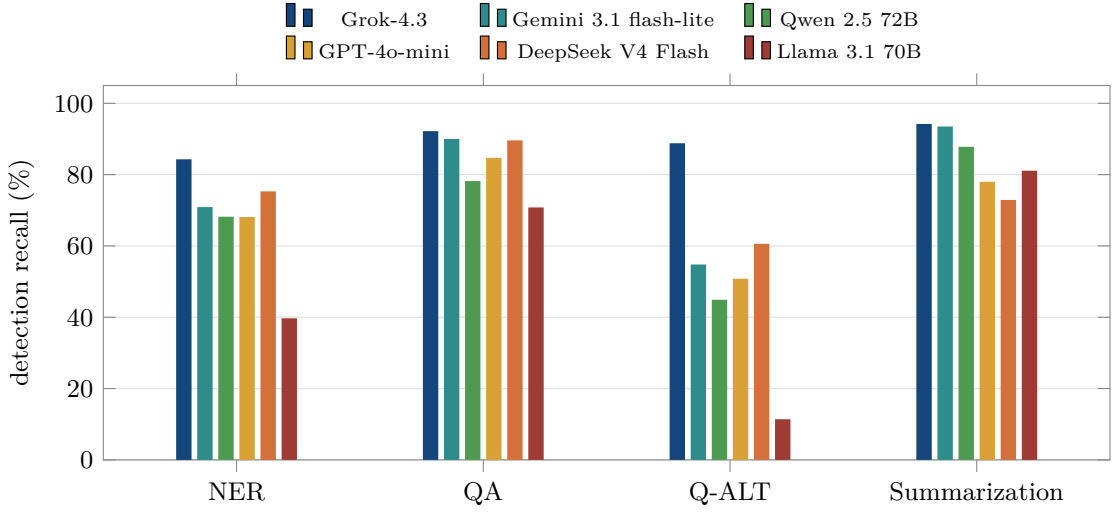


Figure 6.1: Detection recall by track and detector. The NER bar is the mean over its seven entity types.

Two observations hold immediately. The ranking is stable: Grok-4.3 leads all four tracks and Llama 3.1 70B is last on all four, a consistency across very different task formats that is itself a result. And Q-ALT is the hardest track, with four of six detectors below 61%. Against these figures, clean accuracy sits between 94% and 100% for almost every detector and category, which locates the difficulty in detection rather than in reading Bengali.

6.2 Track 1: Entity-Level Detection

Table 6.1 reports detection recall for all seven entity types.

Table 6.1: Track 1 detection recall (%) by entity type, with per-detector and per-category means.

Detector	Age <i>n</i> =532	Dosage <i>n</i> =495	Medicine <i>n</i> =1058	Specialist <i>n</i> =650	Health C. <i>n</i> =577	Med. Proc. <i>n</i> =242	Symptom <i>n</i> =1481	Mean
Grok-4.3	99.2	97.4	94.7	83.5	69.2	73.6	72.5	84.3
DeepSeek V4 Flash	97.4	83.0	79.6	80.2	68.5	51.2	67.2	75.3
Gemini 3.1 flash-lite	97.9	87.3	75.3	64.9	57.0	62.0	51.9	70.9
Qwen 2.5 72B	99.6	81.0	69.9	72.3	67.9	55.8	30.9	68.2
GPT-4o-mini	94.7	82.2	73.3	64.6	61.7	53.7	46.3	68.1
Llama 3.1 70B	73.1	50.9	54.4	15.9	22.7	31.8	28.9	39.7
Category mean	93.7	80.3	74.5	63.6	57.8	54.7	49.6	

Entity type strongly predicts difficulty, answering RQ2 affirmatively. The category means fall monotonically from Age at 93.7% to Symptom at 49.6%, and the division

is interpretable. Age and Dosage are close to solved because the corrupted value is a *number*, and a number that does not fit its context is visible without clinical knowledge. Specialist, Health Condition and Medical Procedure require knowing medicine: substituting a pulmonologist for a cardiologist is wrong only to a reader who knows what those words denote.

Clean accuracy runs between 93.9% and 100% for every detector and category with two exceptions, both on Medicine: DeepSeek V4 Flash at 76.7% and Llama 3.1 70B at 79.8%. DeepSeek is therefore the one detector whose recall on Medicine (79.6%) exceeds its clean accuracy, meaning it over-reports rather than under-reports.

The Llama row deserves separate comment. At 15.9% on Specialist it accepts five of every six swapped entities, while scoring 93.9% on the matching clean sentences. A model that performs near ceiling on one half of a pair and near floor on the other is not failing to read Bengali; it is agreeing with whatever the prompt proposes. The same pattern recurs, more starkly, in Track 3.

6.3 Track 2: Answer-Level Detection

Table 6.2: Track 2 results (%). Wrong-answer recall is the mean over three distractor variants, which agree to within about three points.

Detector	Accepts correct answer	Rejects wrong answer
Grok-4.3	86.3	92.2
Gemini 3.1 flash-lite	86.1	90.0
DeepSeek V4 Flash	81.0	89.6
GPT-4o-mini	66.9	84.7
Qwen 2.5 72B	67.2	78.2
Llama 3.1 70B	71.1	70.8

Table 6.2 reports both halves of the pairing. The measurement is stable: the three distractor variants agree to within roughly three percentage points, so the ranking is not an artefact of which distractor was drawn. More interestingly, the bias direction reverses relative to Track 1. There the models under-flag, accepting corrupted text; here they over-flag. GPT-4o-mini rejects a third of the medically correct answers while still catching 84.7% of the wrong ones, and Qwen behaves the same way. One model therefore carries two opposite failure modes depending only on framing, which is the clearest evidence for building four task formats rather than one.

6.4 Track 3: Question-Level Detection

Track 3 asks a model not to answer but to refuse, and is the hardest track in the benchmark. Table 6.3 breaks the results down by corruption technique.

Table 6.3: Track 3 detection rate (%) by technique. Items: Contradiction 436, Referent Deletion 472, Unrealism 86.

Detector	Contradiction	Unrealism	Referent Deletion	Overall
Grok-4.3	95.0	91.9	82.6	88.8
DeepSeek V4 Flash	72.0	76.7	47.0	60.6
Gemini 3.1 flash-lite	62.4	62.8	46.4	54.8
GPT-4o-mini	46.1	58.1	53.8	50.8
Qwen 2.5 72B	51.8	47.7	37.9	44.9
Llama 3.1 70B	8.3	14.0	13.8	11.4
Mean	55.9	58.5	46.9	

Technique matters as much as model. Referent Deletion is the hardest technique on average, 9 points below Contradiction and 12 below Unrealism, and it is the hardest technique individually for four of the six detectors. The explanation is structural: Contradiction and Unrealism both put something *false* into the question, which sufficient clinical knowledge can catch, whereas Referent Deletion puts nothing false in at all. It removes something, leaving a confident reference with no antecedent, and detecting an absence is a materially different operation from checking what is present.

Llama 3.1 70B selected one of options A–D on 878 of 994 questions in which no such option could be correct. This is the same failure seen in Track 1 in its most extreme form: the model is not evaluating the question against its options but completing the pattern the prompt suggests.

6.5 Track 4: Document-Level Detection

Table 6.4: Track 4 overall results (%) and measured run cost over 2,377 items per detector.

Detector	Clean	Corrupted	Overall	Run cost (USD)
Grok-4.3	89.1	94.2	93.2	4.567
Gemini 3.1 flash-lite	89.3	93.5	92.6	0.137
Qwen 2.5 72B	89.6	87.8	88.1	0.489
GPT-4o-mini	93.6	78.0	81.1	0.098
Llama 3.1 70B	78.0	81.1	80.5	0.697
DeepSeek V4 Flash	89.6	72.9	76.2	0.121

GPT-4o-mini again inverts its Track 1 profile: best of all six on untouched pairs at 93.6%, but only 78.0% on corrupted sources. The cost column answers RQ4 bluntly. Grok-4.3 finishes first at 93.2% but costs \$4.57 against \$0.137 for Gemini 3.1 flash-lite, which finishes 0.58 points behind — a thirty-three-fold cost difference for a gap well inside the noise of a single run. Model scale fares no better as a predictor: the 70B open-weight model places last on every track it completed.

Table 6.5: Track 4 recall (%) by corruption operation. Deletion becomes easier as N rises; Coarsen collapses.

Detector	Del. $N=1$	Del. $N=2$	Del. $N=3$	Substitute	Coarsen
Grok-4.3	98.0	100.0	100.0	95.9	72.7
Gemini 3.1 flash-lite	97.4	100.0	100.0	94.0	71.7
Qwen 2.5 72B	90.6	99.2	99.7	84.3	62.3
Llama 3.1 70B	85.3	92.1	96.0	72.2	59.3
DeepSeek V4 Flash	70.9	82.8	84.8	73.1	51.2
GPT-4o-mini	77.6	91.3	93.3	80.2	42.7
Mean	86.6	94.2	95.6	83.3	60.0

Table 6.5 separates the same items by the operation applied, and two clean gradients answer RQ3. Deletion becomes *easier* as N rises, from 86.6% at one fact to 95.6% at three: more missing material means more signal, and the monotonic trend — which holds for all six detectors individually, without exception — confirms that N works as the intended difficulty dial. Second, Coarsen collapses. At a mean of 60.0% it sits 23 points below Substitute and 36 below Delete $N=3$, and GPT-4o-mini catches only 42.7% of it, worse than chance on a binary task, which means it

is systematically wrong rather than merely uncertain. The reason follows from the operation: the source still discusses the detail and still points the same way, so a detector checking for topical agreement finds it and stops.

Table 6.6 disaggregates the same items by the category of the targeted fact.

Table 6.6: Track 4 recall (%) by targeted fact category, across the six detectors.

Fact category	Best	Worst	Mean
Medicine	98.1	79.1	90.7
Other	98.7	81.1	90.6
Dosage	100.0	68.8	90.3
Health Condition	98.1	78.9	90.2
Age	100.0	76.5	89.6
Symptom	96.7	74.6	85.9
Test Result	87.4	70.4	79.9
Temporal	88.8	66.1	78.1

Temporal and Test Result are the two hardest categories, and they are precisely the categories in which Coarsen applies most naturally: a date or a laboratory value can be made vaguer without being made false, whereas a medicine name largely cannot. The category result and the operation result are therefore two views of the same mechanism.

6.6 Qualitative Examples

Table 6.7 gives one item per corruption type, with the number of detectors that flagged it. These are drawn across the difficulty range rather than selected to favour the benchmark.

Table 6.7: Representative items and how many of the detectors caught each.

Operation	Original	Corrupted	Caught
NER, Medicine	হিস্টাসিন	সিটিরিজিন	3/6
NER, Spe- cialist	“Heart er doctors”	“Pulmonologist”	3/6
NER, Health Cond.	বদহজম	গ্যাস্ট্রাইটিস	3/6
Q-ALT, Un- realism	কত BMI কে স্বাভাবিক ওজন হিসেবে ধরা হয়?	premise fixes the normal range at 20.0–24.9, so no option is correct	1/6
Summ., Coarsen	বার বার নেগেটিভ হচ্ছে	বার বার আশানুরূপ হচ্ছে না	0/6
Summ., Sub- stitute	শ্বাসকষ্ট হয় কিছুটা	মাথা ঘোরে কিছুটা	0/6

The Medicine example is instructive: both terms are antihistamines, so the substitution is clinically plausible but wrong, and it splits the detector panel exactly in half. The Unrealism example is stronger still. The true normal BMI range begins at 18.5, which was option C of the original question; with the premise falsely fixed at 20.0–24.9, no option can be correct, yet five of six detectors selected one anyway.

6.7 Cross-Track Discussion

Five findings hold across all four tracks. *The benchmark measures a real failure:* clean accuracy of 94–100% alongside recall as low as 11% shows the difficulty lies in detection, not in reading Bengali. A detector at ceiling on clean text and near chance on corrupted text is not a weak reader but a reader carrying a prior that the text it is shown is fine — exactly the prior a medical deployment cannot afford. *Failure is structured,* moving predictably with both entity type and operation, which is what makes the benchmark diagnostic rather than merely comparative. *Bias direction is task-dependent,* reversing between Tracks 1 and 2. *Scale and price are weak predictors.* And *absence is harder than error:* the two hardest conditions in the benchmark, Referent Deletion and Coarsen, are both cases in which nothing stated is false.

No prior benchmark measures Bengali medical faithfulness, so direct numerical comparison is not meaningful. Two qualitative points stand. The monotonic relationship between N and accuracy reproduces in Bengali the behaviour BN [4] report for English clinical dialogue, which is evidence that fact-controlled construction transfers across languages and registers. And the Substitute arm shared with FactCC [10] is *not* the hardest condition here — Coarsen is 23 points harder — so a benchmark restricted to substitution would have overstated how well current models handle unfaithfulness.

Chapter 7

Ethical, Social, and Environmental Considerations

Ethical Aspects. The corpora are Bengali patient messages and publicly available examination questions. Because patient messages can carry identifying detail, no personally identifying information is introduced, reconstructed or inferred at any stage; the pipelines edit clinical facts only, and prompts forbid adding information beyond what the edit requires. Prior work whose methods we adapt [1, 4, 10] is cited at the point of adaptation, so that the boundary between borrowed and original method is unambiguous. A second consideration is specific to this project: its output is, by construction, a corpus of plausible-looking but medically wrong Bengali clinical text, which carries misuse potential if mistaken for genuine advice. It is therefore released with every item explicitly labelled, the corrupted span recorded and the operation named. The benchmark is a measuring instrument and must not be used as a source of medical information.

Social Aspects. Hallucination benchmarks are overwhelmingly English, so the assurance that a deployed model behaves safely is far weaker for a Bengali speaker than for an English one, even when the same model serves both. Making Bengali medical faithfulness measurable supplies evidence a deployer would need before putting such a system in front of patients, and the results argue for caution: a model that accepts five of six swapped clinical entities is not fit for unsupervised health advice. That a low-cost model matches a frontier model here also carries an equity implication — adequate safety checking need not be restricted to organisations that can afford frontier inference.

Environmental Aspects. Evaluating six models over roughly 86,700 API calls

has a real energy cost, reduced deliberately. Token caps were set per prompt, in one case to three tokens; checkpointing and resume mean an interrupted run never recomputes completed work; and the faithful pair in Track 4 is deduplicated by document rather than scored once per variant, which removed 8,634 redundant calls. No model was trained or fine-tuned. The cost analysis in Table 6.4 is itself an efficiency result: a flash-tier model within 0.6 points of a frontier model at one thirty-third the cost is evidence for a lower-energy deployment choice.

Chapter 8

Challenges and Limitations

8.1 Challenges

API cost. One frontier detector cost \$4.57 on a single track against \$0.10–0.14 for flash-tier models. This constrained which models could cover which tracks and was the direct cause of the coverage decisions described below.

No annotator pool. With three project members and no external annotators, human verification could not be performed at the scale of the assembled dataset, which is why the verification stage was piloted rather than run in full.

API reliability at scale. Credit exhaustion and rate limiting interrupted long runs constantly; key rotation, checkpointing and repair passes were built specifically to make completion possible.

Reasoning models returning empty completions. A model whose reasoning budget exceeds the token cap returns an empty string rather than an error. Discovered only after a batch had been spent, this led to the mandatory pre-flight probe.

Designing corruptions that hold up. Half the candidate operations did not survive the pilot, leaving 96–100% of the claim verbatim in the source. All three read convincingly by hand; only measuring claim survival exposed them.

8.2 Limitations

Verification not applied to the released set. This is the most significant limitation. The verification judge is implemented and was piloted on 400 items, but it was not run over the full assembled dataset within the timeline, so released items passed only

the two mechanical filters. A residual proportion may carry an incorrect gold label, which would depress all six detectors’ recall roughly equally; the reported *rankings* are therefore more trustworthy than the absolute values.

Detection only, no mitigation. The project measures whether models catch hallucination. It does not attempt to reduce or repair it, so the results describe a problem without yet testing a remedy.

Incomplete coverage. In the summarization track, 7,483 corrupted items in the training split remain unscored.

No clinical validation. All corruptions are model-generated; no clinician has confirmed that the substituted entities are plausible in context, nor that the coarsened values genuinely remove support.

Statistical treatment. Each track uses a single prompt at temperature 0, with no prompt-sensitivity study, self-consistency sampling or significance testing. Differences of one or two points should not be treated as meaningful.

Single generator. Every corruption in the released benchmark was produced by one model. The pilot compared four generators and found detector scores varied little between them, but the possibility of a generator-specific artefact cannot be excluded on that evidence alone.

Chapter 9

Conclusion and Future Work

9.1 Findings and Contributions

RQ1 is answered affirmatively but narrowly: frontier models can detect Bengali medical hallucination, but only Grok-4.3 does so reliably on both halves of a pair. **RQ2** affirmatively: entity type predicts difficulty, with category means falling from 93.5% on Age to 54.7% on Medical Procedure, the division falling cleanly between numeric and knowledge-bound types. **RQ3** affirmatively and more strongly — the operation matters more than the entity type, with mean recall spanning 36 points from Coarsen at 60.0% to three-fact deletion at 95.6%. **RQ4** negatively: a 70B open-weight model places last on every track, and a flash-tier model matches a frontier reasoning model to within 0.6 points at one thirty-third the cost.

The project delivers a hallucination-detection benchmark for Bengali medical text spanning four task formats, 19,396 labelled items and 104,502 detector judgements. It shows that fact-controlled construction transfers from English clinical dialogue to informal Bengali text, contributes the Coarsen operation isolating partial support, and demonstrates a generator/detector separation that removes the self-evaluation loop prior Bengali work identifies in its own design.

9.2 Future Work

1. **Extend to open-ended question answering.** Every task format in this benchmark is closed-form: entity substitution, multiple choice, multiple choice with refusal, and summarization against a fixed source. Free-form generation remains untouched, and is where hallucination is most consequential, since no

option list constrains the model and no reference summary supports a direct comparison. Evaluating it requires claim-level decomposition of the generated answer followed by per-claim adjudication against the source, and is the largest remaining gap in coverage.

2. **Diagnose the causes of failure.** The benchmark reports where detection breaks down but not why. The Coarsen result is the natural starting point: replacing a specific value with a vaguer one that remains true of the original defeats every detector tested, which suggests that models verify consistency rather than sufficiency. Whether that pattern holds across entity types, corruption depth and model scale is answerable in part from the existing data and in full with targeted probes.
3. **Hallucination mitigation.** The natural successor to measuring detection is improving it: chain-of-thought prompting, retrieval support, and an explicit “not sure” option are all testable against this benchmark without new data.
4. **Run the verification judge at scale.** The stage is written and piloted; executing it over the full dataset converts a well-filtered dataset into a verified one, and is a precondition for trusting the absolute values.
5. **Close the coverage gaps.** Score the 7,483 unevaluated summarization items in the training split.
6. **Score span localisation,** which needs no new data since every item already stores the original and replacement spans, and asks a strictly harder question than binary detection.
7. **Human clinical validation** of a stratified sample by Bangladeshi clinicians.
8. **Cheap supervised baselines.** With the benchmark in place, the trained-classifier approach ruled out in section 4.5 becomes available; fine-tuning BanglaBERT [3] would test whether a small supervised model can approach a commercial API.
9. **Beyond binary verdicts:** calibrated confidence and significance testing, so that reported gaps carry error bars.
10. **Other low-resource languages.** The pipeline is language-agnostic apart from its prompts; Hindi, Urdu or Nepali would test whether these failure modes are Bengali-specific.

9.3 Concluding Remarks

The central result is a gap rather than an average. Six contemporary models read faithful Bengali medical text almost perfectly and miss a large share of the corruptions inserted into it, and that gap widens systematically as the corruption becomes less like an error and more like an omission. Coarsening a value rather than falsifying it defeats every detector tested, and no existing benchmark isolates that condition.

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Appendices

Appendix A

Weekly Meeting Summary

Table A.1 records the project meetings held between May and September 2026. Meetings were held roughly weekly, on Wednesdays or Fridays, and were either face-to-face or online; those attended by the supervisor are marked. Two gaps in the schedule correspond to university examination periods, during which no meetings were held. Where the whole team worked on the same activity, most often during the literature review and topic selection phases, the contribution is recorded once across the three member columns.

Table A.1: Group Contribution and Next Plan Table

Sl.	Date	Nashat Islam	Subha Tamanna Mrittika	Atia Zaman	Next Plan
1	1 May with supervisor	Initial consultation on how to select a research area. No domain fixed; the team agreed to survey several fields before committing.			Survey candidate research areas broadly.
2	6 May online	Broad paper survey spanning human-computer interaction, retrieval-augmented generation, natural language processing and applied machine learning. Reading list divided three ways.			Prepare a comparative literature review.
3	15 May with supervisor	Literature review presentation covering the surveyed fields. Supervisor advised comparing the areas on data availability and on whether progress could be measured objectively.			Second reading round against those criteria.
4	22 May	Second reading round. Human-computer interaction and retrieval-augmented generation assessed against natural language processing.			Present the comparison and choose an area.
5	29 May with supervisor	Second literature review presentation. Natural language processing selected as the working area on the strength of available Bengali resources.			Narrow to a sub-area within NLP.
6	3 Jun	Survey of NLP subareas: biomedical NLP, retrieval-augmented generation and low-resource language processing. Three candidates carried forward.			Compare the three for the supervisor.

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Table A.1 continued from previous page

Sl.	Date	Nashat Islam	Subha Tamanna Mrittika	Atia Zaman	Next Plan
7	12 Jun with supervisor	Comparative review of the three subareas. Biomedical NLP favoured; hallucination identified as a recurring unsolved theme across the reviewed work.			Decide the re- search topic.
8	17 Jun with supervisor	Topic fixed: hallucination in large language models. Gap analysis assigned to be carried out over the examination period.			Review the hal- lucination litera- ture.
Midterm examination break, 20 June – 10 July 2026. No meetings held.					
9	15 Jul with supervisor	Hallucination literature reviewed, including Med-HALT, Med-Hallu and CMHE. Gap confirmed: no hallucination resource exists for Bengali medical text.			Survey candidate source datasets.
10	17 Jul	Scope fixed as Bengali medical text. BanglaMedQA, HealthNER and BanglaCHQ-Summ selected as source datasets.			Design the cor- ruption protocol.
11	22 Jul with supervisor	Benchmark design agreed. Source-side corruption adopted: the source document is edited and the reference left untouched, so that faithful and hallucinated items share one text distribution.			Begin implemen- tation.
12	24 Jul	Initialised the repos- itory; assembled the BanglaMedQA corpus and built correct-option res- olution for Track 2.	Reviewed summa- rization corrup- tion operations from the literature.	Prepared the first detector evaluation run.	Report on pilot feasibility.
13	29 Jul online	Opened the API cost ledger and fixed the repository layout.	Drafted the atomic fact extraction schema.	Completed the Llama 3.1 70B and Grok pilot runs.	Freeze Track 2 and open the NER track.
14	5 Aug	Restructured the repository into per- task directories; finalised three inde- pendent distractor variants drawn from real examination options.	Continued the sum- marization corrup- tion design.	Revised the symp- tom sentence set and added datasets of sentences hallu- cinated by several models.	Convert the NER corpus and begin entity-swap gen- eration.
15	12 Aug	Generated hallu- cinated text for the Medicine cat- egory and ran its evaluation.	Specified the opera- tion set for the sum- marization track.	Converted the NER test set from BIO tagging into column format.	Complete the remaining NER categories.
16	19 Aug with supervisor	Ran NER evalua- tion for the Age, Dosage and Medi- cal Procedure cate- gories.	Generated and eval- uated the Health Condition and Spe- cialist categories; introduced the Coarsen operation.	Supported the NER evaluation runs and began Track 3 tech- nique design.	Freeze NER; agree the summa- rization specifica- tion.
17	24 Aug online	Pre-examination checkpoint. NER track frozen. Seeded target selection and the full summarization pipeline specification agreed so that implementation could begin immediately after the examinations.			Implement the summarization pipeline.

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Table A.1 continued from previous page

Sl.	Date	Nashat Islam	Subha Tamanna Mrittika	Atia Zaman	Next Plan
<i>Final examination break, 25 August – 11 September 2026. No meetings held.</i>					
18	12 Sep online	Committed the Track 3 preprocessing notebook and the Q-ALT detector evaluation outputs.	Committed the summarization pipeline, the original and corrupted datasets and the accuracy tables.	Committed raw and per-evaluator outputs for all six detector models and the final accuracy results.	Consolidate results and draft the report.
19	13 Sep with supervisor	Results reviewed. Three corruption operations cut on claim-survival evidence, retaining Delete, Substitute and Coarsen. Report structure approved and the presentation divided three ways.			See below.

Next Plan

The immediate objective beyond this report is to develop the benchmark into a publishable contribution. Four lines of work follow.

First, the existing resource must be consolidated to publication standard. This means closing the remaining coverage gaps, executing the verification judge over the full dataset so that the labels are verified rather than merely filtered, and adding calibrated confidence and significance testing so that reported gaps carry error bars.

Second, the benchmark will be extended to open-ended question answering. Every task format built so far is closed-form, and free-form generation is both untouched and the setting in which hallucination carries the greatest clinical risk.

Third, the project will move from reporting incidence to diagnosing cause, beginning with the finding that corruptions which coarsen a value rather than falsify it defeat every detector tested.

Fourth, mitigation strategies will be evaluated against the benchmark, including chain-of-thought prompting, retrieval grounding and cheap supervised baselines, to establish whether detection can be improved at a cost that would make it deployable rather than only measurable.

Appendix B

Code and Implementation Artifacts

The project repository is available at <https://github.com/nashatislam/benmedh-allu-priv>. All work described in this report is contained in that repository.

B.1 Quantitative Summary: GitHub Metrics

Table B.1 presents the core development contributions for each team member, measured over the full history of the repository between 24 July 2026 and 13 September 2026.

Table B.1: Developer Contribution Metrics

Team Member	Total Commits	Net Δ LoC (Additions)	Net Δ LoC (Deletions)
Nashat Islam	37	412,468	346
Subha Tamanna Mrittika	2	66,532	0
Atia Zaman	16	18,993	1,044
Total	55	497,993	1,390

The two columns measure different things and should be read together. Commit counts reflect commit granularity rather than volume of work: the generated dataset and evaluation files were committed individually by some members and in consolidated batches by others, giving averages of 1,187, 11,148 and 33,266 added lines per commit for Atia Zaman, Nashat Islam and Subha Tamanna Mrittika respectively. Line counts are likewise dominated by generated artefacts, since the cor-

rupted datasets and per-model evaluation outputs are stored as CSV files that are far larger than the notebooks and scripts that produce them.

B.2 Feature Ownership and Technical Contribution

This section details the two most significant technical contributions by each member, as recorded in Table B.2. The literature survey, benchmark design decisions, pilot generator selection, human evaluation and the final report were carried out jointly by all three members.

Table B.2: Feature ownership and technical contribution by team member

Team Member	Technical Summary
Nashat Islam 220041211	
Nashat Islam	Feature: Track 3 (Q-ALT) generation and evaluation. Designed and implemented the three question-corruption techniques, Contradiction, Unrealism and Referent Deletion, which alter the premise of a question so that no listed option can be correct. Added the fifth option that allows a model to decline to answer, generated the full set of 994 altered questions, and ran the detector evaluation over the whole track. This is the hardest track in the benchmark and the one that separates the detectors most sharply.
Nashat Islam	Feature: Track 2 (QA) construction and NER data preparation. Built the option-resolution step that pairs each question with its verified correct answer and with real examination distractors, producing three independent distractor variants rather than model-invented wrong answers. Prepared the NER corpus by converting it from BIO tagging into the entity-based column format the generators consume, and shared the Track 1 detector evaluations. Established the repository architecture and the API cost ledger used throughout the project.
Subha Tamanna Mrittika 220041227	

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Table B.2 continued from previous page

Team Member	Technical Summary
Subha Tamanna Mrittika	Feature: Track 1 (NER) generation and evaluation. Implemented the entity-swap generation that substitutes a same-class entity for the original, covering every entity category except Symptom, and shared the detector evaluations across the track. Handled the two incompatible label polarities between the entity categories so that recall figures remain comparable, and prepared the benchmark datasets used by the generators.
Subha Tamanna Mrittika	Feature: Track 4 (Summarization) preprocessing and generation. Implemented atomic fact extraction across ten clinical categories, converting each source document into the set of checkable claims the corruption operates on. Built the seeded target-selection mechanism that lets the program rather than the model choose which facts to alter, preserving the gold record and making the difficulty dial reproducible. Designed the Coarsen operation, which replaces a specific value with a vaguer one that remains true of the original and so isolates partial support, and implemented the source-corruption pipeline that applies it.
Atia Zaman 220041241	
Atia Zaman	Feature: Track 1 Symptom category generation and evaluation. Generated and evaluated the Symptom entity category, the largest in the track at 1,481 sentences, applying the same-class substitution used across the other categories and running the full detector panel over it. Symptom proved the hardest entity type in the benchmark, with a category mean of 49.6% against 93.7% for Age.

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Table B.2 continued from previous page

Team Member	Technical Summary
Atia Zaman	Feature: Track 2 and Track 4 detector evaluation. Ran the detector panel over the question-answering track, covering both the faithful and the paired-distractor halves for all six models. Carried out the summarization track evaluation across the same six detectors, producing the raw verdict files, the per-evaluator outputs and the final overall accuracy results from which the Track 2 and Track 4 figures are reported.

B.3 Verification: GitHub Screenshots

The following dated screenshots provide verification of the team’s activity and contribution metrics as recorded by the GitHub platform, shown in Figure B.1 and Figure B.2.

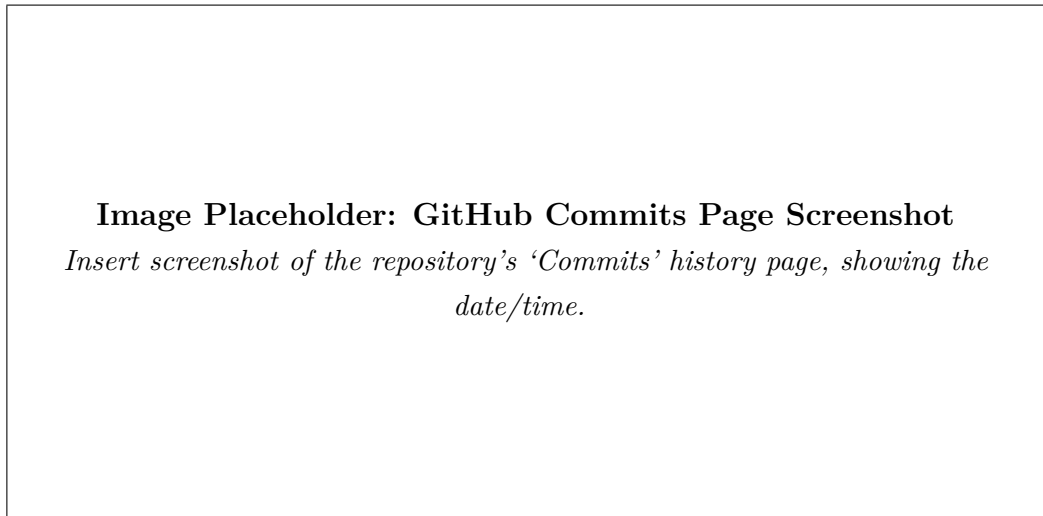


Figure B.1: Screenshot of the GitHub Commit History, verifying date and commit frequency. (Dated: [Insert Date])

Image Placeholder: GitHub Contributors Graph Screenshot

Insert screenshot of the repository's 'Insights → Contributors' graph, verifying contribution distribution.

Figure B.2: Screenshot of the GitHub Contributors Graph, validating individual contributions over time. (*Dated: [Insert Date]*)